

Hodge Deligne Numbers for Hypersurfaces in Toric Varieties

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1 Mike Meeting 03 Aug

Mike took us through some of the basic ideas. It was a long meeting, but there were a lot of ideas, and so this is a writeup for some of the ideas from that meeting.

Any text in purple indicates either a question from myself or an incomplete thought.

1.1 Definition of Hodge-Deligne Numbers and Some Basic Properties

Let X be a variety over $\mathbb{C}$ of dimension¹ n . The *Hodge-Deligne numbers* denoted $e^{p,q}(X)$ for $0 \leq p, q \leq n$ are a collection of integers satisfying properties soon to be described. Moreover the polynomial $e_X(x, y) := \sum_{0 \leq p, q \leq n} e^{p,q}(X) x^p y^q$ is called the *Hodge-Deligne polynomial*. The Hodge-Deligne numbers and polynomial satisfy the following properties

- The Hodge-Deligne polynomial is symmetric in x, y , $e_X(x, y) = e_X(y, x)$. In other words, the Hodge-Deligne numbers are symmetric in p, q , $e^{p,q}(X) = e^{q,p}$.
- If X is reducible then the Hodge-Deligne polynomial decomposes additively over the components: if $X = Y \amalg Z$ (presumably these have to be closed subsets?) then $e_X = e_Y + e_Z$. (A variety is by definition irreducible, so, perhaps the most general setting is for a separated, finite type over $\mathbb{C}$ -scheme? Or maybe this property is supposed to be for closed subvarieties?)
- If X is a product variety then its Hodge-Deligne polynomial decomposes multiplicatively: if $X = Y \times_{\mathbb{C}} Z$ then $e_X = e_Y e_Z$.
- If X is projective and smooth (over $\mathbb{C}$) then recall that the hodge numbers are the dimensions of the $\mathbb{C}$ -vector spaces given by cohomology over the sheaf of differentials

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega_X^p).$$

¹A variety is defined to be separated, irreducible, finite type k -scheme. I am trying to remember why we have that a variety has a well-defined dimension.

In this setting, the Hodge-Deligne numbers are related to the Hodge numbers in the following way

$$e^{p,q}(X) = (-1)^{p+q} \cdot h^{p,q}(X).$$

1.2 Examples

Example 1 (A point). Let $X = \{*\} = \text{Spec}(\mathbb{C})$. We have that X is smooth and projective. The following Hodge calculation will give that $e_X(x, y) = 1$.

Since X is affine $H^q(X, \Omega_X^p)$ is non-vanishing only for $q = 0$. Recall that $\Omega_X^p = 0$ for all $p > 0$ (I have forgotten why this is true, but it follows because X is a point. I think it should be easy to build the module of differentials in this case once one remembers the definition and construction) and so $h^{p,q} = 0$ for all $p, q > 0$. The only possible non-zero Hodge number we may have is for $p = q = 0$. We have the following

$$\begin{aligned} h^{0,0}(X) &:= H^0(X, \Omega_X^0) \\ &= \dim_{\mathbb{C}} H^0(X, \mathcal{O}_X) && \text{By definition} \\ &= \dim_{\mathbb{C}} \Gamma(X, \mathcal{O}_X) && \text{By the cohomology axioms} \\ &= \dim_{\mathbb{C}} \mathbb{C} && X = \text{Spec}(\mathbb{C}) \\ &= 1. \end{aligned}$$

Overall, we have that $h^{0,0}(X) = 1$ and $h^{p,q}(X) = 0$ for all other p, q . The claimed result follows.